Empirical Economics (2024) 67:75–96
https://doi.org/10.1007/s00181-023-02551-2

Forecasting the volatility of European Union allowance futures with macroeconomic variables using the GJR-GARCH-MIDAS model

Huawei Niu[1] · **Tianyu Liu**[1]

Received: 6 February 2023 / Accepted: 22 December 2023 / Published online: 26 January 2024

Abstract
Building on the GJR-GARCH model, this paper uses the mixed-data sampling (MIDAS) approach to link monthly realized volatility of EU carbon future prices and macroeconomic variables to the volatility of EU carbon futures market and proposes the GJR-GARCH-MIDAS model incorporating macroeconomic variables including the economic sentiment indicator of the EU, the harmonized index of consumer prices of the EU, the European economic policy uncertainty index and ECB's marginal lending facility rate (GJR-GARCH-MIDAS-X models). An empirical analysis based on the monthly macroeconomic variables and daily EUA futures data shows that the above four low-frequency macroeconomic variables have significant positive or negative impacts on the long-term volatility of EUA future prices, respectively. The GJR-GARCH-MIDAS-X models significantly outperform other competing models, including the GJR-GARCH model, GARCH-MIDAS model and standard GJR-GARCH-MIDAS model, in terms of out-of-sample volatility forecasting, which suggests that macroeconomic variables contain important information for EUA future price volatility forecasts. In particular, the GJR-GARCH-MIDAS model with harmonized index of consumer prices (HICP) (GJR-GARCH-MIDAS-HICP model) performs best in out-of-sample volatility forecasting, and our findings are robust to different forecasting windows.

Keywords EUA futures · Macroeconomic variables · GJR-GARCH · MIDAS · Volatility forecasting

JEL Classification C32 · C53 · G17

✉ Tianyu Liu
liutianyu@cumt.edu.cn

[1] School of Economics and Management, China University of Mining and Technology, Xuzhou 221116, Jiangsu Province, China

1 Introduction

The European Union Emissions Trading System (EU ETS) was introduced formally on January 1, 2005. After years of development, it has gradually become the largest and most mature carbon emissions trading system in the world. Ye et al. (2021) argue that EU EST is not only for spot trading, but also for derivatives (e.g., futures and options), and that carbon futures, as a commodity, have the function of price discovery, which provide a reference for carbon spot trading. Moreover, carbon futures can effectively circumvent the volatility risk of transaction prices and increase the market liquidity at the same time (Zhou and Li 2019).

Recently, the European Union Allowance (EUA) futures market has developed rapidly, and it is of great practical significance to characterize and measure the fluctuation of EUA futures market prices. The price fluctuations of carbon futures can reflect the overall market price trend, which provides a basis for carbon market decision makers to formulate effective management policies. Moreover, since the risk mechanism for the EU carbon market has not yet been completed, the reasonable and effective volatility forecasting method can help improve the risk management capability, which is important to promote the EU carbon futures market stability (Liu et al. 2021). Therefore, many studies use some models to forecast the volatility of the EUA futures market. Traditionally, volatility models are based on the generalized autoregressive conditional heteroskedasticity (GARCH) model (Bollerslev 1986) and stochastic volatility (SV) model (Taylor 1994). Kim et al. (2017) investigate the volatility of EUA future prices using the SV model and its extensions. Since the parameters of the GARCH-like models can be estimated easily by using the classical maximum likelihood method, the GARCH-like models have gained a great deal of attention from academic researchers. Chevallier (2011) analyzes the conditional variance of EUA future prices during 2005–2008 using the EGARCH model and detects instability in the volatility of EUA future prices based on the retrospective tests and forward-looking tests. Zeitlberger and Brauneis (2016) model EUA spot and future price returns with the GARCH and Markov switching GARCH models. Huang et al. (2021) propose the VMD-GARCH/LSTM-LSTM model for the volatility forecasting of EUA future prices using GARCH model and long short-term memory network, and the empirical results show that the VMD-GARCH/LSTM-LSTM model has better forecasting performance compared with other models. Other studies also model and forecast EUA futures market volatility based on GARCH-like models (Byun and Cho 2013; Zeng et al. 2021).

A large number of studies find that the volatility of financial asset returns is asymmetric (leverage effect), i.e., "bad" news (negative shocks) of equal intensity is more likely to increase the volatility than "good" news (positive shocks). So, taking into account the asymmetric effect in volatility can improve the volatility forecasting ability of the model (Pan and Liu 2018; Guo et al. 2022). However, the GARCH model fails to capture the asymmetric effect in volatility. Then Glosten et al. (1993) propose the GJR-GARCH model to capture the asymmetry of asset return volatility based on the GARCH model. The GJR-GARCH model provides superior volatility forecasts compared to the GARCH model (Kim et al. 2008; Onwukwe et al. 2011). Subsequently, some scholars apply the GJR-GARCH model to the carbon futures market.

Rannou and Barneto (2016) develop a bivariate GJR-GARCH model to capture the asymmetric behavior of volatility of the EUA future (spot) prices. To account for the asymmetry in the volatility of carbon future price returns, Jiao et al. (2018) employ the GJR-GARCH(1, 1) model to conduct the related study.

A growing body of studies suggest that changes in macroeconomic conditions might cause volatility in carbon trading market. And the strong carbon-economy link solidly grounded in economic theory. On the one hand, as with other commodities, the carbon price is determined by the market balance between supply and demand. The demand for carbon allowances largely depends on macroeconomic conditions, while the supply of carbon allowances remains relatively stable and to some extent inelastic to market conditions (Jiao et al. 2018). Hence, economic growth boosts the demand for carbon allowances by companies, which will increase the volatility of carbon future prices. On the other hand, macroeconomic conditions may have an impact on the price of fossil fuels and the demand for fossil fuels, which in turn can affect carbon dioxide emissions and the demand for carbon emission allowances, which in turn may affect carbon future prices. For example, fossil fuel prices rise in the European region, and energy sources that can replace fossil fuels will not increase significantly in the short term. Therefore, some European countries, such as Germany and France, can only restart coal-fired power plants, which will make carbon emissions increase and exacerbate the volatility of carbon future prices. Chevallier (2009) and Bredin and Muckley (2011) investigate the impact of macroeconomic conditions on EUA future prices. Other researchers have also studied in depth the macroeconomic impact on carbon market prices (Jiménez-Rodriguez 2019; Yuan and Yang 2020). However, the above research approaches require that the same sampling frequency of data. Carbon market price data are high-frequency (daily) data, while many macroeconomic variable data are low-frequency (monthly or quarterly) data. The traditional processing method is to convert high-frequency data to low-frequency data or choose other alternative indicators, obviously, the processing method will lose important information and affect the estimates and forecasts of EUA future price volatility, which cannot meet the actual needs.

To solve the problem of modeling data that are sampled at different frequencies, Ghysels et al. (2004) propose the mixed-data sampling (MIDAS) approach. The MIDAS approach can make full use of effective information from the data and has received a great deal of attention from many scholars. Engle et al. (2013) propose the GARCH-MIDAS model by extending the GARCH model with the modeling ideas of the MIDAS model. Moreover, the model decomposes volatility into long-term and short-term components and uses the MIDAS approach to link the long-term components to low-frequency (e.g., monthly) macroeconomic variables, which can effectively enhance the volatility prediction accuracy. Subsequently, many scholars have conducted a series of studies using the GARCH-MIDAS model (Salisu et al. 2022; Mei and Xie 2022; Su et al. 2022). In particular, some studies use the GARCH-MIDAS model to examine the impact of macroeconomic variables on the volatility of stock market and to explore the predictive ability of macroeconomic variables for stock market (Conrad and Kleen 2020; Wei et al. 2021). Moreover, Liu et al. (2021) and Dai et al. (2022) investigate the effect of economic policy uncertainty (EPU) on the volatility in the EU carbon market and its prediction performance. The empirical results show that the GARCH-MIDAS model with EPU outperforms other GARCH-type models in terms of out-of-sample volatility forecasting. However, it is worth pointing out that the above studies only consider the EPU index and do not fully consider the impact of macroeconomic variables on EUA future price volatility and their prediction ability and the GARCH-MIDAS model fails to consider the asymmetry of volatility.

Therefore, building on the GJR-GARCH model, this paper uses the mixed-data sampling (MIDAS) approach to directly link monthly macroeconomic variables to the volatility of the carbon futures market and proposes the GJR-GARCH-MIDAS model incorporating macroeconomic variables including the economic sentiment indicator of the EU, the harmonized index of consumer prices of the EU, the European economic policy uncertainty index and ECB's marginal lending facility rate (GJR-GARCH-MIDAS-X models). In order to test the rationality and superiority of the GJR-GARCH-MIDAS model, we apply the model to the daily EUA future prices and monthly macroeconomic indicators. We compare the GJR-GARCH-MIDAS-X models with other competing models, including the GJR-GARCH model, GARCH-MIDAS model and standard GJR-GARCH-MIDAS model, in terms of both in-sample fitting and out-of-sample forecast.

Taken together, the contributions of the paper are twofold. Firstly, we introduce the GJR-GARCH-MIDAS model combing the GJR-GARCH and GARCH-MIDAS model. Compared to the GJR-GARCH model, the GJR-GARCH-MIDAS model allows macroeconomic variables directly enter the specification of the long-term component of volatility. Consequently, using the GJR-GARCH-MIDAS model that incorporates macroeconomic variables, we are able to study the relation between macroeconomic variables and the volatility of EUA future prices. Compared to the GARCH-MIDAS model, the GJR-GARCH-MIDAS model could capture the asymmetry of EUA future price volatility. Secondly, the empirical results show that the GJR-GARCH-MIDAS-X models outperform the standard GJR-GARCH-MIDAS model in terms of both in-sample fitting and out-of-sample forecast. The low-frequency macroeconomic variables have significant positive or negative impacts on the long-term volatility of EUA future prices, respectively. Moreover, macroeconomic variables can greatly improve the forecast accuracy for EUA future price volatility. In particular, the GJR-GARCH-MIDAS-HICP model performs best in out-of-sample volatility forecasting, and our findings are robust to different forecasting windows.

The remainder of the paper is organized as follows. In Sect. 2, we introduce the models used. Section 3 presents the empirical results, while Sect. 4 concludes the paper.

2 The model

2.1 The GJR-GARCH model

Glosten et al. (1993) propose the GJR-GARCH model to capture the asymmetry of asset return volatility based on the GARCH model. GJR-GARCH(1, 1) is presented as follows:

$$r_i = \mu + \epsilon_i, \quad \epsilon_i = \sigma_i z_i, \quad z_i \sim N(0, 1), \tag{1}$$
$$\sigma_i^2 = \omega + (\alpha_1 + \alpha_2 I_{\{\epsilon_{i-1}<0\}})\epsilon_{i-1}^2 + \beta\sigma_{i-1}^2, \tag{2}$$

where r_i is the asset return at day i and μ is the conditional mean of the asset return. Since the conditional mean is usually small, it is generally assumed to be 0 in the empirical analysis. ϵ_i is the return innovation at day i and z_i is the standardized return innovation, which is assumed to obey the standard normal distribution with independent identical distribution. σ_i^2 is the conditional variance of the asset return, where $I_{\{\epsilon_{i-1}<0\}}$ is the characteristic function; when ϵ_{i-1} is less than zero, the value of I is one, otherwise, I is zero.

2.2 The GARCH-MIDAS model

To allow data with different sampling frequencies in the model, Engle et al. (2013) propose the GARCH-MIDAS model by combining the GARCH model with the MIDAS approach. The GARCH-MIDAS model decomposes volatility into two components: a short-run (high-frequency) component and a long-run (low-frequency) component. In order to distinguish between different frequencies, another subscript t is introduced. $r_{i,t}$ represents the asset return on day i of the period (month) t. Thus, the new equation can be formulated as

$$r_{i,t} = \mu + \sigma_{i,t} z_{i,t}, \quad \forall i = 1, ..., N_t, \tag{3}$$

where N_t is the number of trading days in period t. $z_{i,t}|\Phi_{i-1,t} \sim i.i.d.N(0, 1)$, given the information set $\Phi_{i-1,t}$ up to day $i-1$ of period t. Decomposing the conditional variance $\sigma_{i,t}^2$ into two components, which can be expressed as follows:

$$\sigma_{i,t}^2 = \tau_t g_{i,t}, \tag{4}$$

where τ_t is the long-run component and $g_{i,t}$ is the short-run component. The short-run component, $g_{i,t}$, is specified as a GARCH(1, 1)-like process, which can be written as

$$g_{i,t} = (1 - \alpha_1 - \beta) + \alpha_1 \frac{(r_{i-1,t} - \mu)^2}{\tau_t} + \beta g_{i-1,t}. \tag{5}$$

To ensure nonnegativity and stationarity for the short-run component $g_{i,t}$, we assume $\alpha_1 \geq 0$, $\beta \geq 0$ and $\alpha_1 + \beta < 1$. The long-run component, τ_t, is modeled in the spirit of the MIDAS regression, which is driven by the smoothing monthly realized volatility (RV) with the weighting scheme $\varphi_k(\gamma_1)$, where γ_1 is a parameter to be estimated and the value of γ_1 can be any real number. The specific forms of τ_t are as follows:

$$\log(\tau_t) = m + \theta_1 \sum_{k=1}^{K} \varphi_k(\gamma_1) \log(\mathrm{RV}_{t-k}), \tag{6}$$

$$\mathrm{RV}_t = \sum_{i=1}^{N_t} r_{i,t}^2, \tag{7}$$

where θ_1 is the coefficient of the long-run effect of RV on volatility and K is the number of MIDAS lags with $\sum_{k=1}^{K} \varphi_k(\gamma_1) = 1$. The weight function is given by

$$\varphi_k(\gamma_1) = \frac{(1 - k/K)^{\gamma_1 - 1}}{\sum_{j=1}^{K}(1 - j/K)^{\gamma_1 - 1}}. \tag{8}$$

Equations (3)–(8) together form the GARCH-MIDAS model.

2.3 The GJR-GARCH-MIDAS model

In this paper, the series of macroeconomic variables are monthly data and the series of carbon future prices are daily data. If the traditional modeling methods with the same frequency data are used, the high-frequency effective information in carbon future prices may be lost, making the parameter estimation and volatility prediction biased. Therefore, we introduce the GJR-GARCH-MIDAS model by combining the GJR-GARCH model, which can capture the asymmetry of asset return volatility, with the GARCH-MIDAS model. The GJR-GARCH-MIDAS model can be written as

$$r_{i,t} = \mu + \sigma_{i,t} z_{i,t}, \quad \forall i = 1, ..., N_t, \tag{9}$$

$$\sigma_{i,t}^2 = \tau_t g_{i,t}, \tag{10}$$

$$g_{i,t} = (1 - \alpha_1 - \beta - \frac{\alpha_2}{2}) + (\alpha_1 + \alpha_2 I_{\{r_{i-1,t}<0\}}) \frac{(r_{i-1,t} - \mu)^2}{\tau_t} + \beta g_{i-1,t}, \tag{11}$$

$$\log(\tau_t) = m + \theta_1 \sum_{k=1}^{K} \varphi_k(\gamma_1) \log(\mathrm{RV}_{t-k}), \tag{12}$$

$$\mathrm{RV}_t = \sum_{i=1}^{N_t} r_{i,t}^2. \tag{13}$$

It can be seen that Eq. (11) is different from Eq. (5). $g_{i,t}$ is specified as a GJR-GARCH(1, 1)-like process. Where $I_{\{r_{i-1,t}<0\}}$ is the characteristic function. When $r_{i-1,t}$ is less than zero, the value of I is one, otherwise, I is zero. Additionally, we assume that $\alpha_1 \geq 0$, $\alpha_2 \geq 0$, $\beta \geq 0$ and $\alpha_1 + \beta + \frac{\alpha_2}{2} < 1$. The coefficient, α_2, can reflect the asymmetric impact of negative and positive shocks on carbon futures price volatility, and if α_2 is significantly greater than 0, it indicates that negative shocks have a greater impact on volatility than positive shocks.

2.4 The GJR-GARCH-MIDAS-X models

The standard GJR-GARCH-MIDAS model is able to portray the time-varying characteristics of long-term volatility, but it does not take into account the impact of macroeconomic factors on volatility. As mentioned in the introduction, macroeconomic conditions may have an impact on the volatility of carbon future prices. Therefore, a reasonable volatility model needs to fully consider the time-varying nature of long-term volatility in financial markets as well as macroeconomic information. So, this study constructs the GJR-GARCH-MIDAS model that incorporates macroeconomic variables (GJR-GARCH-MIDAS-X models). In the models, Eq. (12) is rewritten as

$$\log(\tau_t) = m + \theta_1 \sum_{k=1}^{K} \varphi_k(\gamma_1) \log(RV_{t-k}) + \theta_2 \sum_{k=1}^{K} \varphi_k(\gamma_2) \log(X_{t-k}), \tag{14}$$

where $\varphi_k(\gamma_2)$ is the Beta-type weight function and it has a similar structure to equation (8). X denotes the macroeconomic variables. θ_1 and θ_2 are the coefficients of the long-run effects of RV and X on volatility, respectively. When θ_2 equals 0, the GJR-GARCH-MIDAS-X models reduce to the standard GJR-GARCH-MIDAS model.

2.5 Maximum likelihood estimation

The GJR-GARCH-MIDAS(-X) models are easy to estimate. We can use the maximum likelihood method to estimate the parameters of the GJR-GARCH-MIDAS(-X) models.[1] Since $z_{i,t}|\Phi_{i-1,t} \sim i.i.d.N(0, 1)$, we have

$$r_{i,t} - \mu = \sigma_{i,t} z_{i,t}, \quad \sigma_{i,t} z_{i,t}|\Phi_{i-1,t} \sim N(0, \sigma_{i,t}^2). \tag{15}$$

Then, the probability density function of $r_{i,t}$ can be written as

$$f(r_{i,t}; \Theta) = \frac{1}{\sqrt{2\pi\sigma_{i,t}^2}} \exp(-\frac{(r_{i,t} - \mu)^2}{2\sigma_{i,t}^2}), \tag{16}$$

where $\sigma_{i,t}^2 = \tau_t g_{i,t}$. Therefore, the log-likelihood function of the GJR-GARCH-MIDAS(-X) models can be written as

$$\ell(r; \Theta) = -1/2 \sum_{t=1}^{T} \sum_{i=1}^{N_t} \left[\log(2\pi) + \log(\tau_t g_{i,t}) + \frac{(r_{i,t} - \mu)^2}{\tau_t g_{i,t}} \right], \tag{17}$$

[1] In this paper, we use MATLAB (R2018a) software to do this, and the related algorithm and code will be made available if the reader require them.

where $r = (r_{1,1}, ..., r_{N_1,1}, ..., r_{1,T}, ..., r_{N_T,T})'$, and Θ denotes all the parameters that are to be estimated in the models.[2] Specifically, we have $\Theta = (m, \alpha_1, \alpha_2, \beta, \theta_1, \gamma_1)'$ for the standard GJR-GARCH-MIDAS model, and $\Theta = (m, \alpha_1, \alpha_2, \beta, \theta_1, \theta_2, \gamma_1, \gamma_2)'$ for the GJR-GARCH-MIDAS-X models. Next, the maximum likelihood estimate of the model parameters could be obtained by maximizing the above log-likelihood function:

$$\hat{\Theta} = arg \max_{\Theta} \ell(r; \Theta). \tag{18}$$

3 Empirical results

3.1 Selection of variables

This paper investigates the impacts of macroeconomic variables on carbon future price volatility and the predictive ability of macroeconomic variables for carbon futures market based on the GJR-GARCH-MIDAS-X models. Before the empirical analysis, the selection of the main variables, such as carbon future prices and macroeconomic indicators, is introduced. The EU carbon trading market is currently the largest in the world, so the paper takes the EU carbon futures market as the research object and selects the daily trading settlement price of European Union Allowance (EUA) futures as the representative of the EU carbon futures market.

Numerous studies have found that macroeconomic conditions can have an impact on carbon trading prices. Therefore, the paper considers the macroeconomic effects on EU carbon future price volatility. There are many indicators that can reflect the macroeconomic conditions, such as GDP, industrial output, and so on. However, when selecting indicators, we should consider not only the representativeness of indicators, but also the comprehensiveness of indicators. Based on the research of other scholars, this paper refines the macroeconomic factors into the following four categories.

Firstly, the condition of national economic development. The available indicators are GDP, economic sentiment indicator, industrial output, income, etc. In this paper, the economic sentiment indicator (ESI) is selected by referring to the studies of Mera et al. (2020); Liudmila and Tamara (2021). We have access to obtain monthly data for ESI, while we only have access to obtain quarterly and annual data for GDP, and GDP data information exhibits less timeliness. Moreover, ESI can portray the changing trend of macroeconomic fluctuations more effectively than industrial output. The EU ESI (EESI) is calculated based on five market confidence indices, and a higher ESI indicates a better macroeconomic development and a more prosperous overall economy.

Secondly, the condition of consumption. The consumer price index (CPI) is one of the most commonly used indicators to reflect the price level and consumption situation in the economy. The harmonized index of consumer prices (HICP) is a good indicator of measuring the overall CPI level in the EU. Therefore, we choose the indicator in the paper.

[2] Note that Zakoian (1994) and Engle et al. (2013) provide the log-likelihood functions for the GJR-GARCH and GARCH-MIDAS models, respectively.

Thirdly, economic policy uncertainty. To measure the degree of economic policy uncertainty, Baker et al. (2016) use textual analysis to develop an economic policy uncertainty (EPU) index for the world's major economies based on the frequency of keywords on newspaper coverage such as economic condition, policy adjustments, and uncertainty. A great deal of studies have found that EPU has a significant impact on the volatility of financial assets. However, there is less literature exploring the relationship between EPU and carbon futures market. Therefore, we want to further explore whether the European EPU (EEPU) will have the impact on the volatility of the EU carbon futures market.

Finally, interest rates. Interest rates are the price of money and have an important orientation to macroeconomics and resource allocation. Furthermore, interest rates are an important lever in the economic system and are the most commonly used policy tool in macroeconomic regulation. There are many types of interest rates in the European financial market, and this paper selects the marginal lending facility rate (MLFR), one of the benchmark interest rates of the European Central Bank, as a proxy variable for European interest rates (Fecht and Weber 2022).

3.2 Data

The data used for this study consist of two components, including daily EU carbon future settlement prices and monthly macroeconomic indicators. The EU ETS is still in its infancy and the market is not yet mature in the first phase (2005–2007). So, the data start from the second phase. The EUA futures data cover the period from January 2, 2008, to May 31, 2022, including 3712 trading days of data. The data are obtained from Wind Database of China.

We consider four macroeconomic indicators (EESI, HICP, EEPU index and MLFR) to study the effect of macroeconomic variables on carbon future price volatility and their predictive ability. The monthly macroeconomic data cover the period from January 2008 to May 2022, which match the sampling phase with the daily EUA futures data, including 173 monthly data. The EESI, HICP and MLFR data can be obtained from Wind Database of China, and EEPU index data are available from https://www.policyuncertainty.com/. The time series plots of EUA futures returns and macroeconomic variables are given in Figs. 1 and 2, respectively. The volatility of EUA futures returns exhibits clear aggregation characteristics as can be seen in Fig. 1. The EU economy experiences severe downturn, affected by the global financial crisis in 2008 and the coronavirus (COVID-19) pandemic, which causes an decrease in EESI. HICP shows an overall upward trend, especially during the COVID-19 pandemic. Major events such as the global financial crisis in 2008, the European debt crisis in 2010, Brexit and the COVID-19 pandemic have led to an increase in the EEPU index. In order to promote economic development, MLFR levels are generally on a downward trend.

Table 1 presents the descriptive statistics for daily returns of EUA futures and monthly macroeconomic variables. As seen in Table 1, daily returns ($r_{i,t}$) and monthly EESI (EESI_t) are negatively skewed, while monthly HICP (HICP_t), monthly EEPU index (EEPU_t) and MLFR (MLFR_t) are positively skewed. All sequences, except

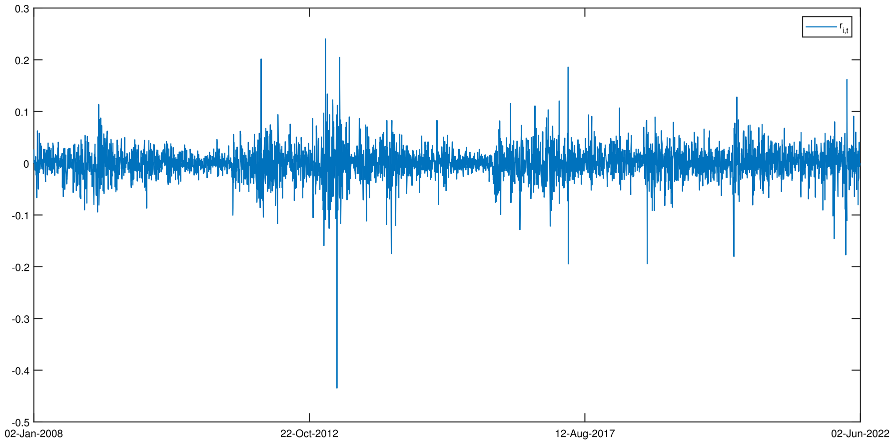

Fig. 1 Daily EUA futures returns

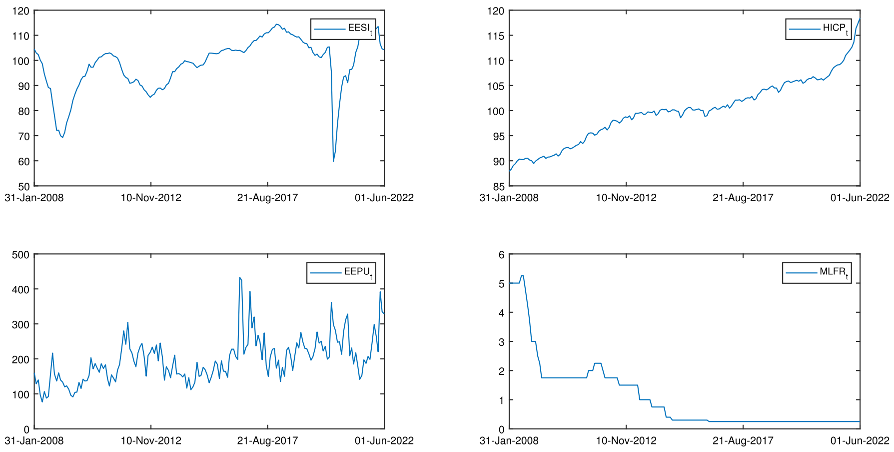

Fig. 2 Monthly macroeconomic variables

HICP$_t$, exhibit the sharp peak and thick tail and non-normal distribution characteristic. Additionally, the Ljung-Box Q statistics for autocorrelation up to 12 lags provide evidence of high persistence (or long-memory property) for the volatility of EUA future prices.

3.3 Estimation results

By adopting the maximum likelihood method, the parameters of the GJR-GARCH-MIDAS-X models can be estimated. Furthermore, in order to test the applicability of the GJR-GARCH-MIDAS-X models in carbon future price volatility forecasting applications and further analyze the effect of macroeconomic variables on carbon future price volatility, we estimate the parameters of the three competing models

Table 1 Descriptive statistics

	Mean	Min	Max	Std	Skewness	Kurtosis	Jarque-Bera	Q(12)
$r_{i,t}$	0.0003	−0.4347	0.2405	0.0316	−0.7786	17.4582	32697.8385	33.9829
$EESI_t$	98.7572	59.8000	116.1000	10.8325	−0.9610	4.1205	35.6768	744.8278
$HICP_t$	99.9194	87.9200	118.4400	6.2668	0.1948	2.8882	1.1848	1446.3098
$EEPU_t$	197.6778	76.8234	433.2775	64.0192	0.9727	4.5796	45.2692	346.8403
$MLFR_t$	1.0873	0.2500	5.2500	1.2317	1.8515	6.2082	173.0375	1147.4201

Q(12) is the Ljung-Box statistic for autocorrelation up to 12 lags including GJR-GARCH model, GARCH-MIDAS model and GJR-GARCH-MIDAS model. Estimation results are reported in Table 2.[3] It is important to note that the maximum lag order K needs to be determined when estimating models that include the MIDAS structure. Conrad and Loch (2015) find that the data will identify the optimal weighting scheme as long as the chosen K is sufficiently large. We choose $K = 12$ for the monthly RV and macroeconomic variables, i.e., the lag time range of the effect of RV and macroeconomic variables on the volatility of the carbon futures market is assumed to be 12 months.

As shown in Table 2, the estimate of the persistence coefficient $\alpha_1 + \frac{\alpha_2}{2} + \beta$ in the GJR-GARCH model is 0.9907, which is very close to 1, exhibiting very high volatility persistence. It is interesting that the estimation result of the persistence coefficient $\alpha_1 + \beta$ for the short-term component is 0.9213 in the GARCH-MIDAS model and the estimates of the persistence coefficient $\alpha_1 + \frac{\alpha_2}{2} + \beta$ for the short-term component of the GJR-GARCH-MIDAS(-X) models range from 0.90 to 0.92. The volatility persistence coefficients of these models are lower than those of the GJR-GARCH model, indicating that the MIDAS structure can capture well the long-term trend component of EUA future price volatility (long memory). In the GJR-GARCH and GJR-GARCH-MIDAS(-X) models, the leverage parameter estimates, α_2, are all positive and statistically significant, indicating a significant asymmetric effect on the volatility of the EUA futures market, i.e., negative return shocks have a greater impact on volatility than positive return shocks.

Moreover, the estimates of the parameter θ_1 are all significantly positive, suggesting that the monthly RV has the significant positive effect on the long-term volatility of EUA future prices, that is, an increase in RV predicts higher levels of the long-term volatility of EUA future prices. Similarly, the estimates of θ_2 in the GJR-GARCH-MIDAS-X models are all statistically significant, suggesting that the monthly macroeconomic variables have significant effects on the long-term components of the EU carbon futures market volatility. Specifically, the estimates of the parameter θ_2 are significantly positive, indicating that the low-frequency macroeconomic variables have the significant positive impact on the long-term volatility of EUA future prices. An increase in EESI, suggesting that the economy is in a period of high growth, leads to an increase in the emissions of carbon dioxide, which could increase the volatility of carbon future prices. An increase in HICP could be due to the increase in the price of fossil energy sources (e.g., coal and oil), which could affect the prices of carbon futures and make them more volatile. And the increase in EEPU could affect the production behavior and carbon emissions of regulated companies, which will also increase the volatility of carbon future prices. Meanwhile, the estimate of the parameter θ_2 in the GJR-GARCH-MIDAS-MLFR model is significantly negative, indicating the negative effect. A decrease in MLFR will lead to lower financing costs for companies, promoting development and increasing carbon emissions, which in turn could increase the volatility of carbon future prices. It is interesting to note that HICP has the largest absolute value of θ_2 among all macroeconomic variables, which means that

[3] To simplify the model structure and improve the out-of-sample forecast of the model, we make the simplifying assumption $\mu = 0$ from Hansen and Huang (2016).

HICP contributes more than other variables in interpreting long-term carbon emission price volatility.

Next, turning to the parameters γ_1 and γ_2, we find that the estimates of the parameters γ_1 and γ_2 are greater than 1 for all related models except the GJR-GARCH-MIDAS-EESI model, which shows that the weight coefficients of the lagged terms of the RV and macroeconomic variables gradually decrease with the increase of lag time, i.e., the closer the time to the current time, the greater the impact of the RV and macroeconomic variables on the volatility of the carbon futures market. In particular, the estimate of γ_2 in the GJR-GARCH-MIDAS-HICP model is larger than those in all other models; thus, the rate of decrease of the former is the fastest.

Finally, we also calculate the Log-lik (log-likelihood function) and the AIC (Akaike information criterion) for all models. The results are given in Table 2. We find that the log-likelihood values of the GJR-GARCH-MIDAS-X models are larger than those of the GJR-GARCH, GARCH-MIDAS and GJR-GARCH-MIDAS models, and the AIC values are smaller than those of the GJR-GARCH, GARCH-MIDAS and GJR-GARCH-MIDAS models, indicating that the GJR-GARCH-MIDAS-X models fit the data better. The results suggest that taking into account asymmetric effect in volatility and macroeconomic information can improve the data fit. It is worth noting that comparing the GJR-GARCH-MIDAS-X models with each other, it can be seen that the GJR-GARCH-MIDAS-HICP model fits the data best in terms of the values of the log-likelihood and AIC.

3.4 Out-of-sample results

In this section, we examine the effectiveness of the GJR-GARCH-MIDAS model for out-of-sample prediction of EUA future price volatility and empirically investigate the out-of-sample predictive ability of macroeconomic variables, which is of importance to market participants. We choose evaluation criteria (loss functions and MCS test) to conduct a comprehensive test of the out-of-sample volatility forecasting ability of the models. We split the sample into an in-sample estimation period (January 2, 2008, to December 31, 2019), including 3088 trading days of data, and an out-of-sample forecast evaluation period (January 2, 2020, to May 31, 2022), including 624 trading days of data. First, we adopt a rolling window scheme with a fixed window size of 3088 for estimating the models and computing the one-day-ahead out-of-sample forecasts. The first forecast date is January 2, 2020. The estimated sample length is then held constant, the window of the estimated model is rolled forward one day to re-estimate the model, and the re-estimated model is used for volatility forecasting. The whole process is repeated until May 31, 2022.

To assess the forecast performance of the models, we use two loss functions, the mean-squared error (MSE) and the quasilikelihood (QLIKE). Patton (2011) shows that they are robust in the sense that they yield the same ranking of two volatility forecasts when using an observed (unbiased) volatility proxy instead of the unobserved true

Table 2 Estimation results

Models	GJR-GARCH	GARCH -MIDAS	GJR-GARCH -MIDAS	GJR-GARCH-MIDAS-X			
				EESI	HICP	EEPU	MLFR
$m\ (\omega)$	0.0000***	−4.3294***	−4.2021***	−9.3112***	−16.5086***	−6.1993***	−4.3302***
	(0.0000)	(0.0872)	(0.0719)	(0.1068)	(0.1408)	(0.0722)	(0.0875)
α_1	0.1078***	0.1546***	0.1015***	0.1012***	0.0964***	0.0995***	0.0969***
	(0.0089)	(0.0089)	(0.0090)	(0.0087)	(0.0092)	(0.0089)	(0.0095)
α_2	0.0458***		0.1092***	0.1065***	0.1179***	0.1225***	0.1178***
	(0.0113)		(0.0138)	(0.0148)	(0.0135)	(0.0147)	(0.0130)
β	0.8600***	0.7667***	0.7522***	0.7585***	0.7630***	0.7463***	0.7576***
	(0.0072)	(0.0139)	(0.0142)	(0.0142)	(0.0125)	(0.0135)	(0.0128)
θ_1		0.6226***	0.6561***	0.6542***	0.5831***	0.6085***	0.6343***
		(0.0187)	(0.0169)	(0.0115)	(0.0061)	(0.0091)	(0.0163)
θ_2				1.1136***	2.6095***	0.3453***	−0.0960***
				(0.0147)	(0.0309)	(0.0150)	(0.0176)
γ_1		5.1411***	5.1982***	4.7308***	4.7498***	5.0735***	4.9961***
		(0.0981)	(0.1095)	(0.0656)	(0.0684)	(0.0905)	(0.0978)
γ_2				0.9396***	48.7519***	5.8143***	7.5158***
				(0.0150)	(0.5682)	(0.1849)	(1.5641)
Log-lik	8115.0207	8123.9288	8135.5568	8138.3685	8141.4342	8138.8324	8137.7702
AIC	−16222.0413	−16237.8576	−16259.1137	−16260.7371	−16266.8684	−16261.6649	−16259.5405

The number in parenthesis is the standard error. *** indicates statistical significance at 1%

volatility. The two loss functions are given by

$$\mathrm{MSE} = \frac{1}{L}\sum_{i=1}^{L}(\sigma_{i+1,t}^{2} - \sigma_{i+1,t|i}^{2})^{2}, \tag{19}$$

$$\mathrm{QLIKE} = \frac{1}{L}\sum_{i=1}^{L}\left(\log(\sigma_{i+1,t|i}^{2}) + \frac{\sigma_{i+1,t}^{2}}{\sigma_{i+1,t|i}^{2}}\right), \tag{20}$$

where $\sigma_{i+1,t|i}^{2}$ and $\sigma_{i+1,t}^{2}$ denote the forecasted volatility obtained from different models and the true volatility, respectively. Since the true volatility is not directly observable, it is necessary to select a "proxy" for the true volatility. We use the square value of the daily real return as a proxy variable for true volatility (Wei et al. 2017; Liu et al. 2021).

Table 3 shows the results of the forecast evaluation on the MSE and QLIKE loss functions. It can be seen from the table that the loss function values of the GJR-GARCH-MIDAS(-X) models are smaller than those of the GJR-GARCH and GARCH-MIDAS models, suggesting that the GJR-GARCH-MIDAS(-X) models outperform both the GJR-GARCH and GARCH-MIDAS models for the volatility forecasts. Therefore, considering the impact of macroeconomic variables on the long-run volatility (MIDAS structure) and the asymmetric effect in volatility plays an important role in volatility forecasting. Furthermore, the extended GJR-GARCH-MIDAS models (GJR-GARCH-MIDAS-EESI, GJR-GARCH-MIDAS-HICP, GJR-GARCH-MIDAS-EEPU and GJR-GARCH-MIDAS-MLFR) outperform the standard GJR-GARCH-MIDAS model, implying that macroeconomic variables contain important information for EUA future volatility forecasting. In particular, the GJR-GARCH-MIDAS-HICP model consistently gives the lowest loss values and significantly outperforms all other models. This may be due to the fact that energy prices are an important component of the HICP, and changes in the HICP may be attributed to the volatility of the energy market, which can affect the volatility of carbon futures market (Liu and Chen 2013; Dutta 2019). Therefore, the HICP might contain more important information about carbon future price volatility forecasts than other macroeconomic factors.

Although the loss function values can reflect the predictive ability of the models, the conclusions cannot be made on the basis of a single loss function. To solve this problem, Hansen (2005) proposes the superior predictive ability (SPA) test, which requires subjective selection of a benchmark model and has certain algorithmic drawbacks. Thus, Hansen et al. (2011) propose the model confidence set (MCS) test based on the loss function and point out that the results obtained from the MCS test are more robust than those obtained from the SPA test. There are a large number of studies that use MCS test to compare the predictive power of models (Koopman et al. 2016; Gong and Lin 2017; Alexander et al. 2018; Cui and Feng 2020; Liu et al. 2022).

The MCS procedure examines a given set of competing models and identifies a set of the best performing models with some confidence level, namely, the MCS. To be specific, it sets a model collection $\mathcal{M}_0$, which contains the volatility forecasting models to be inspected and the corresponding loss values are then calculated from the out-of-sample volatility forecasts of each model according to the loss functions

Table 3 Out-of-sample forecast evaluation results

Models	GJR-GARCH	GARCH -MIDAS	GJR-GARCH -MIDAS	GJR-GARCH-MIDAS-X			
				EESI	HICP	EEPU	MLFR
MSE	7.1681e−06	7.1314e−06	6.9519e−06	6.8906e−06	**6.8823e−06**	6.9433e−06	6.8930e−06
QLIKE	−6.0046	−6.0132	−6.0338	−6.0376	**−6.0405**	−6.0400	−6.0349

(MSE and QLIKE), denoted as $L_{i,j,m}$ for model j of the loss function i (out-of-sample window m=1,..., M). Thus, for the two forecasting models u and v in model collection $\mathcal{M}_0$, the relative loss function values are denoted as

$$d_{i,uv,m} = L_{i,u,m} - L_{i,v,m}, u, v \in \mathcal{M}_0. \tag{21}$$

The set of superior objects is then defined as $\mathcal{M}^*$, which can be expressed as following:

$$\mathcal{M}^* \equiv \{u \in \mathcal{M}_0 : E(d_{i,uv,m}) \leq 0, for\,all\; v \in \mathcal{M}_0\}. \tag{22}$$

The MCS procedure is conducted by a series of significance tests in the model collection $\mathcal{M}_0$. The models with relatively poor predictive ability are eliminated until no models are excluded from $\mathcal{M}_0$. The null hypothesis assumes that any two models have the same predictive power, which is given by

$$H_{0,\mathcal{M}} : E(d_{i,uv,m}) = 0, \text{ for all } u, v \in \mathcal{M}_0. \tag{23}$$

The MCS procedure is based on an equivalence test, $\delta_{\mathcal{M}}$ and elimination rule, $e_{\mathcal{M}}$. The equivalence test, $\delta_{\mathcal{M}}$, is used to test the hypothesis H_0 for any $\mathcal{M} \subset \mathcal{M}_0$, and $e_{\mathcal{M}}$ identifies the object of $\mathcal{M}$ that is to be removed from $\mathcal{M}$, in the event that H_0 is rejected. After repeating the two tests, we obtain the set, which consists of the set of "surviving" objects and is referred to as the model confidence set (MCS). Moreover, the MCS test statistics include range statistic (T_{R}) and semi-quadratic statistic(T_{SQ}), which are defined as

$$T_{\mathrm{R}} = \max_{u,v \in \mathcal{M}} \frac{|\bar{d}_{i,uv}|}{\sqrt{\operatorname{var}(\bar{d}_{i,uv})}}, \tag{24}$$

$$T_{\mathrm{SQ}} = \max_{u,v \in \mathcal{M}} \frac{(\bar{d}_{i,uv})^2}{\operatorname{var}(\bar{d}_{i,uv})}, \tag{25}$$

$$\bar{d}_{i,uv} = \frac{1}{M} \sum_{m=H+1}^{H+M} d_{i,uv,m}, \tag{26}$$

where $\bar{d}_{i,uv}$ is the average loss difference. In the paper, the significance level is equal to 0.1. If the p-value is larger than 0.1, the corresponding model is a "surviving" model, implying that its forecasting performance is relative good, otherwise, it means that its prediction performance is poor. And the larger the p-value, the higher the prediction accuracy of the model. Since the asymptotic distributions of the statistics T_{R} and T_{SQ} depend on the "nuisance parameters", the bootstrap method based on 10,000 replications is used to obtain the corresponding p-value.

Table 4 presents the MCS test results of the models. It can be seen from the table that the GARCH-MIDAS model and the GJR-GARCH-MIDAS(-X) models pass the MCS test in all cases, indicating that the models with MIDAS structure provide better performance than the GJR-GARCH model. Most of the GJR-GARCH-MIDAS-X models

Table 4 MCS test results

	Statistics	GJR-GARCH	GARCH -MIDAS	GJR-GARCH -MIDAS	GJR-GARCH-MIDAS-X EESI	HICP	EEPU	MLFR
MSE	T_R	0.0915	**0.1970**	**0.1970**	**0.9123**	**1.0000**	**0.1970**	**0.9123**
	T_{SQ}	0.0667	**0.1222**	**0.1222**	**0.9101**	**1.0000**	**0.1838**	**0.9101**
QLIKE	T_R	0.0174	**0.2600**	**0.4919**	**0.8759**	**1.0000**	**0.9283**	**0.7104**
	T_{SQ}	0.0326	**0.2157**	**0.5869**	**0.8552**	**1.0000**	**0.9283**	**0.6910**

Bold entries indicate the model is included in the MCS at a 10% significance level

Table 5 MCS test results for forecast window of 500

	Statistics	GJR-GARCH	GARCH -MIDAS	GJR-GARCH -MIDAS	GJR-GARCH-MIDAS-X EESI	HICP	EEPU	MLFR
MSE	T_R	**0.2648**	**0.4614**	**0.6809**	**0.8390**	**1.0000**	**0.8390**	**0.8390**
	T_{SQ}	**0.3351**	**0.5750**	**0.7423**	**0.7556**	**1.0000**	**0.7556**	**0.7556**
QLIKE	T_R	0.0587	**0.9965**	**0.9227**	**0.9965**	**0.9965**	**0.9965**	**1.0000**
	T_{SQ}	**0.1152**	**0.9959**	**0.9817**	**0.9959**	**0.9959**	**0.9959**	**1.0000**

Bold entries indicate the model is included in the MCS at a 10% significance level

Table 6 MCS test results for forecast window of 1000

	Statistics	GJR-GARCH	GARCH -MIDAS	GJR-GARCH -MIDAS	GJR-GARCH-MIDAS-X EESI	HICP	EEPU	MLFR
MSE	T_R	0.0248	**0.1367**	**0.4497**	**0.4497**	**1.0000**	**0.4497**	**0.4497**
	T_{SQ}	**0.1553**	**0.4107**	**0.7366**	**0.7366**	**1.0000**	**0.7366**	**0.7366**
QLIKE	T_R	0.0437	**0.3000**	**0.5580**	**0.9600**	**1.0000**	**0.9600**	**0.9600**
	T_{SQ}	0.0942	**0.4373**	**0.8023**	**0.9517**	**1.0000**	**0.9517**	**0.9517**

Bold entries indicate the model is included in the MCS at a 10% significance level

have larger MCS test p-values than the GJR-GARCH-MIDAS model, especially the GJR-GARCH-MIDAS-HICP model not only passes the MCS test in all cases, but also obtains the largest MCS test p-value ($p = 1$). Such results suggest that macroeconomic variables, especially HICP, play an important role in explaining the volatility of EUA future prices and improving the accuracy of volatility forecasts.

3.5 Robustness checks

To empirically examine the robustness of the out-of-sample prediction results and illustrate the superior volatility forecasting ability of the GJR-GARCH-MIDAS-HICP model, we choose two different forecast windows (out-of-sample periods), 500 and 1000, to re-run the MCS tests. The out-of-sample forecast evaluation results are presented in Tables 5 and 6, respectively.

As can be seen in Tables 5 and 6, the MCS test results are largely consistent with those in Table 4, and the GJR-GARCH-MIDAS-HICP model has the largest MCS test p-value ($p = 1$) in most cases. It shows that considering the effect of HICP on the volatility of EUA future prices helps to forecast volatility more accurately, and the GJR-GARCH-MIDAS-HICP model has superior volatility forecasting ability. The findings are robust with respect to different forecasting windows.

4 Conclusions

Numerous studies find that macroeconomic conditions have the impact on carbon markets, but few studies examine carbon future price volatility from the perspective of the external macro environment. In view of this, building on the GJR-GARCH model, this paper uses the mixed-data sampling (MIDAS) approach to directly link monthly realized volatility and macroeconomic variables to the volatility of the carbon futures market and proposes the GJR-GARCH-MIDAS model incorporating macroeconomic variables (GJR-GARCH-MIDAS-X models). Using monthly macroeconomic variables (EESI, HICP, EEPU and MLFR) and daily EU carbon futures data as research samples, we empirically investigate the impact of macroeconomic variables on EUA future price volatility and explore the predictive ability of macroeconomic variables for carbon futures market. Our study has led to the following three conclusions.

Firstly, the above four low-frequency macroeconomic variables have significant positive or negative impacts on the long-term volatility of EUA future prices, respectively, i.e., the increase in EESI, HICP and EEPU and the decrease in MLFR will exacerbate the long-term volatility of EUA future prices.

Secondly, the GJR-GARCH-MIDAS-X models outperform the GJR-GARCH, GARCH-MIDAS and GJR-GARCH-MIDAS models both in-sample fitting and out-of-sample volatility forecasts. This shows that considering the asymmetric effect in volatility and the impact of macroeconomic variables on EU carbon future price volatility in the model can significantly improve the fitting effect and the forecasting ability of the model.

Finally, comparing the volatility forecasting ability of the GJR-GARCH-MIDAS-X models incorporating different macroeconomic variables (EESI, HICP, EEPU and MLFR), it is found that the GJR-GARCH-MIDAS-HICP model provides the most superior volatility forecasts and the finding is robust to different forecasting windows.

The research results of the paper could provide some valuable implications for the improvement of the EU ETS. Firstly, when formulating carbon market policies, policy makers should consider the impact of macroeconomic variables on carbon market price volatility and their predictive role and take a comprehensive and long-term view of the policies. For example, policymakers must strive to maintain the continuity and stability of EU carbon market policies and reduce economic policy uncertainty in order to achieve the goal of promoting the healthy and orderly development of the EU carbon market. Secondly, the market participants could forecast the European carbon market volatility based on changes in macroeconomic variables (e.g., HICP), especially for emitting companies that need to participate in trading to achieve compliance in the EU ETS. In conclusion, both policymakers and market participants could manage risk by predicting volatility of the European carbon market.

The paper provides a novel perspective and methodology for modeling and forecasting the volatility of carbon futures market, which is of great theoretical and practical importance. Of course, it is also worthwhile as a future research endeavor to investigate the performance of the GJR-GARCH-MIDAS(-X) models in the interest rate market and foreign exchange market. It is worth pointing out that applying the model to carbon option pricing is also an important research direction for the future.

Funding This work was supported by the National Natural Science Foundation of China (No. 71871120) and the Fundamental Research Funds for the Central Universities (No. 2022ZDPYSK02) and the Ministry of Education of Humanities and Social Science Project of China (No. 23YJCZH006).

Data availability The EESI, HICP, MLFR and EUA data could be obtained from Wind Database of China, And EEPU index data are available from https://www.policyuncertainty.com/.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

References

Alexander C, Kaeck A, Sumawong A (2018) A parsimonious parametric model for generating margin requirements for futures. Eur J Oper Res 273(1):31–43
Baker SR, Bloom N, Davis SJ (2016) Measuring economic policy uncertainty. Q J Econ 131(4):1593–1636
Bollerslev T (1986) Generalized autoregressive conditional heteroskedasticity. J Econom 31(3):307–327
Bredin D, Muckley C (2011) An emerging equilibrium in the EU emissions trading scheme. Energy Econ 33(2):353–362
Byun SJ, Cho HJ (2013) Forecasting carbon futures volatility using GARCH models with energy volatilities. Energy Econ 40:207–221
Chevallier J (2009) Carbon futures and macroeconomic risk factors: a view from the EU ETS. Energy Econ 31(4):614–625
Chevallier J (2011) Detecting instability in the volatility of carbon prices. Energy Econ 33(1):99–110
Conrad C, Kleen O (2020) Two are better than one: volatility forecasting using multiplicative component GARCH-MIDAS models. J Appl Econom 35(1):19–45
Conrad C, Loch K (2015) Anticipating long-term stock market volatility. J Appl Econom 30:1090–1114
Cui Y, Feng Y (2020) Composite hedge and utility maximization for optimal futures hedging. Int Rev Econ Financ 68:15–32
Dai PF, Xiong X, Duc HTL, Wang JQ (2022) The impact of economic policy uncertainties on the volatility of European carbon market. J Commod Mark 26:100208
Dutta A (2019) Impact of carbon emission trading on the European Union biodiesel feedstock market. Biomass Bioenerg 128:105328
Engle RF, Ghysels E, Sohn B (2013) Stock market volatility and macroeconomic fundamentals. Rev Econ Stat 95(3):776–797
Fecht F, Weber P (2022) Private value of central bank liquidity and Banks' bidding behavior in variable rate tender auctions. J Bank Financ 136:106221

Ghysels E, Santa-Clara P, Valkanov R (2004) The MIDAS touch: mixed data sampling regression models. CIRANO, Montreal
Glosten LR, Jagannathan R, Runkle DE (1993) On the relation between the expected value and the volatility of the nominal excess return on stocks. J Financ 48(5):1779–1801
Gong X, Lin BQ (2017) Forecasting the good and bad uncertainties of crude oil prices using a HAR framework. Energy Econ 67:315–327
Guo XZ, Huang YS, Liang C, Umar M (2022) Forecasting volatility of EUA futures: new evidence. Energy Econ 110:106021
Hansen PR (2005) A test for superior predictive ability. J Bus Econ Stat 23(4):365–380
Hansen PR, Huang Z (2016) Exponential GARCH modeling with realized measures of volatility. J Bus Econ Stat 34(2):269–287
Hansen PR, Lunde A, Nason JM (2011) The model confidence set. Econometrica 79(2):453–497
Huang YM, Dai XY, Wang QW, Zhou DQ (2021) A hybrid model for carbon price forecasting using GARCH and long short-term memory network. Appl Energy 285:116485
Jiao L, Liao Y, Zhou Q (2018) Predicting carbon market risk using information from macroeconomic fundamentals. Energy Econ 73:212–227
Jiménez-Rodriguez R (2019) What happens to the relationship between EU allowances prices and stock market indices in Europe? Energy Econ 81:13–24
Kim, Jong K, Seok H (2008) A study on the characteristics of asymmetric volatility by industry in Korean stock market. Korean J Bus Adm 21(6):2947–2964
Kim J, Park YJ, Ryu D (2017) Stochastic volatility of the futures prices of emission allowances: Bayesian approach. Phys A 465:714–724
Koopman SJ, Lucas A, Scharth M (2016) Predicting time-varying parameters with parameter-driven and observation-driven models. Rev Econ Stat 98(1):97–110
Liu HH, Chen YC (2013) A study on the volatility spillovers, long memory effects and interactions between carbon and energy markets: the impacts of extreme weather. Econ Model 35:840–855
Liu J, Zhang ZT, Yan LZ, Wen FH (2021) Forecasting the volatility of EUA futures with economic policy uncertainty using the GARCH-MIDAS model. Financ Innov 7(1):76
Liu YY, Niu ZB, Suleman MT, Yin LB, Zhang HW (2022) Forecasting the volatility of crude oil futures: The role of oil investor attention and its regime switching characteristics under a high-frequency framework. Energy 238:121779
Liudmila K, Tamara L (2021) The relationship of economic sentiment and GDP growth in Russia in light of the Covid-19 crisis. Entrep Bus Econ Rev 9(1):7–29
Mei DX, Xie YT (2022) U.S. grain commodity futures price volatility: Does trade policy uncertainty matter? Financ Res Lett 48:103028
Mera VI, Silaghi MIP, Turcu C (2020) Economic sentiments and money demand stability in the CEECs. Open Econ Rev 31(4):343–369
Onwukwe CE, Bassey BEE, Isaac IO (2011) On modeling the volatility of nigerian stock returns using GARCH models. J Math Res 3(4):31
Pan ZY, Liu L (2018) Forecasting stock return volatility: a comparison between the roles of short-term and long-term leverage effects. Phys A 492:168–180
Patton A (2011) Volatility forecast comparison using imperfect volatility proxies. J Econ 160(1):246–256
Rannou Y, Barneto P (2016) Futures trading with information asymmetry and OTC predominance: another look at the volume/volatility relations in the European carbon markets. Energy Econ 53:159–174
Salisu AA, Gupta R, Demirer R (2022) Global financial cycle and the predictability of oil market volatility: evidence from a GARCH-MIDAS model. Energy Econ 108:105934
Su YD, Liang C, Zhang L, Zeng Q (2022) Uncover the response of the U.S grain commodity market on El Nino-Southern oscillation. Int Rev Econ Financ 81:98–112
Taylor SJ (1994) Modeling stochastic volatility: a review and comparative study. Math Financ 4:183–204
Wei Y, Liu J, Lai XD, Hu Y (2017) Which determinant is the most informative in forecasting crude oil market volatility: Fundamental, speculation, or uncertainty? Energy Econ 68:141–150
Wei Y, Bai L, Yang K, Wei GW (2021) Are industry-level indicators more helpful to forecast industrial stock volatility? Evidence from Chinese manufacturing purchasing managers index. J Forecast 40(1):17–39
Ye SQ, Dai PF, Nguyen HT, Huynh NQA (2021) Is the cross-correlation of EU carbon market price with policy uncertainty really being? A multiscale multifractal perspective. J Environ Manag 298:113490
Yuan NN, Yang L (2020) Asymmetric risk spillover between financial market uncertainty and the carbon market: a GAS-DCS-copula approach. J Clean Prod 259:120750

Zakoian JM (1994) Threshold heteroskedastic models. J Econ Dyn Control 18(5):931–955
Zeitlberger ACM, Brauneis A (2016) Modeling carbon spot and futures price returns with GARCH and Markov switching GARCH models: evidence from the first commitment period (2008–2012). Cent Europ J Oper Res 24(1):149–176
Zeng SH, Jia JM, Su B, Jiang CX, Zeng GW (2021) The volatility spillover effect of the European Union (EU) carbon financial market. J Clean Prod 282:124394
Zhou K, Li YW (2019) Carbon finance and carbon market in China: progress and challenges. J Clean Prod 214:536–549